FUCHENG TIAN

Computational Mechanics | Fracture Mechanics | Soft Materials

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Public CV - updated August 2026

RESEARCH PROFILE

My research integrates computational mechanics, nonlinear fracture, and soft-matter physics. I develop numerical methods and physics-informed models to reveal mechanisms of damage, dynamic instability, and toughening in nonlinear materials, especially soft solids.

Research interests: computational mechanics; fracture, damage and toughening of soft materials; dynamic fracture instabilities; phase-field methods; peridynamics; double-network materials.

PROFESSIONAL EXPERIENCE

2026.07 - Present	**Postdoctoral Researcher** Computational Granular Mechanics Lab, Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology | Advisor: Prof. Jidong Zhao
2025.04 - 2026.07	**Postdoctoral Researcher** Faculty of Advanced Life Science, Hokkaido University, Japan | Co-advisor: Prof. Jian Ping Gong
2023.04 - 2025.03	**JSPS Postdoctoral Fellow** Faculty of Advanced Life Science, Hokkaido University, Japan | Host: Prof. Jian Ping Gong
2023.01 - 2023.03	**Postdoctoral Researcher** Faculty of Advanced Life Science, Hokkaido University, Japan | Co-advisor: Prof. Jian Ping Gong
2020.09 - 2022.12	**Postdoctoral Researcher** School of Engineering Science, University of Science and Technology of China | Co-advisor: Prof. Yong Ni

EDUCATION

2015.09 - 2020.06	**Ph.D., Nuclear Science and Technology** National Synchrotron Radiation Laboratory, University of Science and Technology of China | Supervisor: Prof. Liangbin Li
2011.09 - 2015.06	**B.Eng., Engineering Mechanics** School of Mechanics and Engineering Science, Zhengzhou University

MAJOR RESEARCH CONTRIBUTIONS

1. Computational mechanics for nonlinear fracture
Developed phase-field, smoothed finite-element, mixed, adaptive, and peridynamic formulations for fracture under large deformation. The methods address mesh distortion, incompressibility, interface fracture, crack branching, and nonlocal failure while improving robustness and computational efficiency.

2. Dynamic fracture instability
Identified an energy-based crack-branching criterion, clarified the origins of limiting crack speed, and established mechanisms for sub-Rayleigh, oscillatory, and supershear fracture. Recent work shows how constitutive nonlinearity and local material scales regulate crack-speed selection.

3. Double-network materials and toughening
Established minimal mechanics-based frameworks for strength-toughness synergy in soft-hard composites and double networks. The models connect composition, load transfer, defect screening, localized-to-delocalized damage, stable necking, and the brittle-to-ductile transition.

4. Viscoelastic flow instability
Developed predictive finite-element models for non-isothermal film casting, oblique stretching, and concentration-driven dual-mode draw resonance, linking constitutive physics to processing-window design.

SELF-DEVELOPED SOFTWARE

- **LHS-DN v1.0-1.2:** GPU-enabled finite-element lattice-network model for mesoscopic damage, fracture, and double-network mechanics.
- **ha-PFM v1.0-3.2:** Explicit/implicit 2D and 3D fracture solvers with adaptive meshing, parallel assembly, FEM/S-FEM discretization, and compressible/incompressible finite-strain formulations.
- **PolyLab v1.0-2.0:** Multiphysics finite-element solver for steady and transient viscoelastic flows, perturbation analysis, and flow stability.

SELECTED PUBLICATIONS

1. **Fucheng Tian**, Feixue Lu, Katsuhiko Sato, Liangbin Li, Bin Li, Jian Ping Gong. Defect screening and load transfer in minimal hard-soft double networks. *arXiv:2605.11481, 2026*.
2. Jiale Ji, **Fucheng Tian**, Shuyu Chen, Kunpeng Cui, Liangbin Li. Dynamic cracks in strips: revealing local effects in brittle fracture. *Engineering Fracture Mechanics 331, 111696, 2026*.
3. **Fucheng Tian**, Katsuhiko Sato, Yong Zheng, Feixue Lu, Jian Ping Gong. Fundamental toughening landscape in soft-hard composites: insights from a minimal framework. *Proceedings of the National Academy of Sciences 122, e2506071122, 2025*.
4. **Fucheng Tian**, Jian Ping Gong. Nonlinearity tunes crack dynamics in soft materials. *Journal of the Mechanics and Physics of Solids 202, 106191, 2025*.
5. Mengnan Zhang, Shuyu Chen, Jiale Ji, Kunpeng Cui, **Fucheng Tian**, Liangbin Li. Mechanisms governing crack speed in peridynamic model. *Engineering Fracture Mechanics 305, 110201, 2024*.
6. Cui Nie, Mengnan Zhang, Fan Peng, Jun Zeng, Kunpeng Cui, **Fucheng Tian**, Liangbin Li. Dual-mode draw resonance instability regulated by concentration fluctuation in polymer solution casting. *Journal of Rheology 68(6), 973-983, 2024*.
7. Mengnan Zhang, Erjie Yang, Cui Nie, Jun Zeng, **Fucheng Tian**, Liangbin Li. An extended ordinary state-based peridynamic model for nonlinear deformation and fracture. *Computer Methods in Applied Mechanics and Engineering 412, 116100, 2023*.
8. Jun Zeng, Jiale Ji, Shuyu Chen, **Fucheng Tian**. Sub-Rayleigh to supershear transition of dynamic mode-II cracks. *International Journal of Engineering Science 188, 103862, 2023*.
9. Shuyu Chen, Jun Zeng, Mengnan Zhang, Jiale Ji, Liangbin Li, **Fucheng Tian**. Fracture of soft materials with interfaces: phase field modeling based on hybrid ES-FEM/FEM. *Engineering Fracture Mechanics 276, 108892, 2022*.
10. **Fucheng Tian**, Mengnan Zhang, Jun Zeng, Bin Li, Liangbin Li. Adaptive stabilized mixed formulation for phase field fracture modeling of nearly incompressible finite elasticity. *International Journal of Mechanical Sciences 236, 107753, 2022*.
11. Jiale Ji, Mengnan Zhang, Jun Zeng, **Fucheng Tian**. Uncovering the intrinsic deficiencies of phase-field modeling for dynamic fracture. *International Journal of Solids and Structures 256, 111961, 2022*.
12. **Fucheng Tian**, Jun Zeng, Mengnan Zhang, Liangbin Li. Mixed displacement-pressure-phase field framework for finite strain fracture of nearly incompressible hyperelastic materials. *Computer Methods in Applied Mechanics and Engineering 394, 114933, 2022*.
13. Jun Zeng, Mengnan Zhang, Erjie Yang, **Fucheng Tian**, Liangbin Li. A tracking strategy for multi-branched crack tips in phase-field modeling of dynamic fractures. *International Journal for Numerical Methods in Engineering 123(3), 844-865, 2022*.
14. Mengnan Zhang, Erjie Yang, Jun Zeng, Jiale Ji, **Fucheng Tian**, Liangbin Li. Numerical study on oblique stretching of viscoelastic polymer film. *Journal of Non-Newtonian Fluid Mechanics 295, 104597, 2021*.
15. **Fucheng Tian**, Xiaoliang Tang, Tingyu Xu, Liangbin Li. An adaptive edge-based smoothed finite element method for phase-field modeling of fractures at large deformations. *Computer Methods in Applied Mechanics and Engineering 372, 113376, 2020*.
16. **Fucheng Tian**, Jun Zeng, Xiaoliang Tang, Tingyu Xu, Liangbin Li. A dynamic phase field model with no attenuation of wave speed for rapid fracture instability in hyperelastic materials. *International Journal of Solids and Structures 202, 685-698, 2020*.
17. **Fucheng Tian**, Xiaoliang Tang, Tingyu Xu, Junsheng Yang, Liangbin Li. Bifurcation criterion and the origin of limit crack velocity in dynamic brittle fracture. *International Journal of Fracture 224, 117-131, 2020*.
18. **Fucheng Tian**, Xiaoliang Tang, Tingyu Xu, Junsheng Yang, Liangbin Li. A hybrid adaptive finite element phase-field method for quasi-static and dynamic brittle fracture. *International Journal for Numerical Methods in Engineering 120(9), 1108-1125, 2019*.
19. **Fucheng Tian**, Xiaoliang Tang, Tingyu Xu, Junsheng Yang, Chun Xie, Liangbin Li. Nonlinear stability and dynamics of nonisothermal film casting. *Journal of Rheology 62(1), 49-61, 2018*.

GRANTS AND PROJECTS

- **National Natural Science Foundation of China, Young Scientists Fund**, "Study on the Mechanism of Subsonic Fracture in Hyperelastic Materials," 2022.01 - 2023.12, RMB 200,000, Principal Investigator (completed).

- **Japan Society for the Promotion of Science Postdoctoral Fellowship**, "Dynamic Fracture Behavior of Double-Network Gels under Large Deformation," 2023.04 - 2025.03, JPY 2,000,000, Principal Investigator (completed).

HONORS AND AWARDS

- National Scholarship for Graduate Students, 2017
- Outstanding Prize, Zhou Peiyuan National Mechanics Competition, 2012
- National Encouragement Scholarship, 2011
- Excellent Student Award, Zhengzhou University, 2012, 2013, 2014